Fiscal Year:	FY 2018	**Task Last Updated:**	FY 08/22/2018
PI Name:	Mohraz, Ali Ph.D.		
Project Title:	Understanding the Morphology and Stability of Bijels Using Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Irvine	**State:**	CA
Zip Code:	92697-2575	**Congressional District:**	45
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	10/01/2013	**End Date:**	08/05/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 8/5/2018 when continuation grant was awarded per D. Chao/GRC and NSSC information (Ed., 1/7/2019) NOTE: End date changed to 12/31/2020 for now per D. Chao/GRC; however, a continuation grant will be awarded (Ed., 7/20/18) NOTE: End date changed to 9/30/2018 per NSSC information (Ed., 12/7/17)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	NNX13AQ69G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Bicontinuous interfacially jammed emulsion gels (bijels) are a new class of soft materials with interpenetrating co-continuous arrangement of two immiscible liquid phases separated by a monolayer of colloidal particles. Because of this unique morphology, bijels offer significant potential for the design and synthesis of composite materials with co-continuous microstructure for energy and human health applications, with initial efforts along this realm already underway. To realize their full technological utility, a more thorough understanding of the physicochemical and morphological factors that influence the mechanical stability and processability of bijels is required. However, these materials are inherently comprised of multiple phases with disparate densities and complex interfacial interactions, leading to complicated physics during their processing, especially in the presence of gravity. In fact, ground-based experiments have already shown deviations from the expected morphological behavior in these systems. Using synergistic ground-based and microgravity experiments and theoretical calculations, the proposed program aims to develop quantitative metrics for the assessment of bijels stability in the presence of external forces such as gravity and surface tension gradients during bijel processing. Success of the proposed program would pave the way for the design and synthesis of novel co-continuous microstructured composites for diverse applications in advanced manufacturing, energy, nanotechnology, and health care. NOTE (Ed., 1/7/2019): Project is continued by "New Bijel Chemistries for Long-duration Microgravity and Ground-based Experiments," grant 80NSSC18K1554 and Dr. Mohraz continues as Principal Investigator.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Bicontinuous interfacially jammed emulsion gels (bijels) have a unique microstructure that is suitable for the development of advanced materials and composites that require many sample-spanning phases to co-exist. For example, next generation battery electrodes are envisioned as 3D arrangements of ion- and electron-conducting phases in coexistence with electrolyte channels that percolate throughout the entire electrode volume. This morphology can be uniquely realized in bijels, as they inherently enjoy a co-continuous microstructure that is self-assembled through an external stimulus (temperature change). Similarly, the co-continuous morphology of bijel-derived materials can prove beneficial for tissue engineering and cell delivery scaffolds or any other structure that requires microbial growth, such as electrodes in microbial fuel cells. Finally, the open pore morphology and uniform domain size uniquely realized in bijels can lead to better hydraulic permeability for a given porosity, leading to enhanced transport and lower energy consumption for fluid flow. Therefore, bijel processing will provide a unique route to advanced energy and health care materials, and understanding the rheology and processability of bijels, which is the focus of this project, will pave the way for the realization of this end goal. Applications of bijels in each of these areas are currently being investigated in our laboratory.
Task Progress:	EDITOR'S NOTE: Project is continued by "New Bijel Chemistries for Long-duration Microgravity and Ground-based Experiments," grant 80NSSC18K1554 and Dr. Mohraz continues as Principal Investigator. REPORT SUBMITTED OCTOBER 2018 During the fifth year of the award, we have made progress on four related fronts: 1) We have expanded the chemical library of bijels to include binary fluid mixtures that are safe, and in the case of accidental leakage, will not pose a health hazard to astronauts, and fluid mixtures that are nonvolatile, to allow long-duration rheological experiments without problems arising from evaporation. This nontrivial task required an exhaustive scan through the library of liquid pairs that may exhibit a miscibility gap under conditions accessible in the laboratory, experimental construction of their phase diagram, and careful tuning of colloidal surface chemistries to achieve neutral wetting conditions with respect to these fluids. Our efforts have been successful, and our next phase of research will involve at least two more bijel chemistries that meet the requirements for handling at the International Space Station (ISS), or for long-duration rheology experiments without evaporation. 2) We have begun a systematic experimental study to address the issue of shelf life and evolving particle chemistry in samples that are prepared in the laboratory for microgravity research several months later. Our strategy has been based on a proposed overshoot method, where the particle wettability is initially rendered highly off-neutral. Upon several months of exposure to the binary liquid mixture, the particles then gradually attain neutral wetting characteristics. Our systematic experiments have identified a particular surface treatment that has shown promise for achieving the desired shelf life. 3) We continued our comprehensive study of the unique morphological aspects of bijels and microporous materials derived from them in the context of multiphase transport through porous media. We hypothesized that the uniform pore diameter and continuous pore phase in a bijel results in higher hydraulic permeability, and also helps improve transport kinetics through these porous materials. To test our hypothesis, we performed a comprehensive study to quantify the salient morphological characteristics of microporous bijel-derived materials and their role in mediating fluid flow through the material. The project involved synthesis, 3D scanning, and computational geometry on bijels and three other classes of commonly used microporous materials (commercially available porous metals, emulsion-templated random porous structures, and inverse opals). Our continued efforts in the reporting period completed this study, resulting in the first draft of a menuscript that will be edited and submitted for publication within the next reporting period. 4) We began a detailed study of the use of bijel-derived materials in human health applications. During the reporting period, two aspects of bijel-derived biomaterials were studied. First, in a study completed and already published in ACS Biomaterials Science and Engineering during the reporting period, we demonstrated the use of composite bijel templated porous hydrogels as cell delivery vehicles that are mechanically robust and offer pore channels that are uniform in diameter, allowing facile cell migration and nutrient transport throughout the material. Second, in an ongoing study on which a PCT patent application has been filed during the reporting period, and the results of which will be submitted for publication in the next reporting period, we have demonstrated the potential ability of bijel-derived materials to enable better tissue integration and minimize foreign body response when used as coatings for implants. ANNUAL REPORT SUBMITTED AUGUST 2017 During the fourth year of the award, we have made progress on three related fronts (these have been broken up into four sections in the submitted report): 1) We have continued our investigation of the rheology of multi-phase emulsions in which mechanical stability is

provided by colloidal particles residing at fluid interfaces. Such materials present a simple model for bijels due to their simple microstructure. Understanding their rheology would be a first step in the study of bijel rheology where the microstructure and the morphology of the interface are much more complicated.

Our efforts in year four have been focused on the rheology of solid-stabilized emulsions over a wide region of their tertiary state diagram. Effects of droplet packing on the rheology of Pickering emulsions, in a wide range of droplet volume fraction from concentrated emulsions to bi-liquid foams (high internal phase emulsions) have been investigated, with three particle classes that represent different strengths of interparticle interactions through the continuous organic phase. Such studies are important steps in understanding the rheology of multiphase mixtures in which particles are sequestered at liquid interfaces, and form a percolating network that imparts gel-like viscoelasticity to the mixture. We utilize a model experimental system which consists of aqueous droplets (65:35 volumetric mixture of dimethyl sulfoxide and water) in a dodecane continuous phase. Droplets are stabilized with either 650 nm hexamethyldisilazane surface-modified silica particles or poly(12-hydroxystearic acid)-grafted poly(methyl methacrylate) (PHSA-PMMA) particles of two different sizes. These mixtures show an intriguing dependence of the zero-shear elastic modulus on droplet volume fraction, which we rationalize by the added excluded volume of particles and the interfacial energy associated with solid-stabilized droplets. Currently we are also performing Langmuir trough experiments on each particle class to better rationalize the data in the context of interfacial elasticity and interparticle interactions at the interface. These results will be submitted to a peer-reviewed journal in the upcoming period. Next step in this endeavor is to resolve the microstructural origins of gel-like rheology in bijels.

2) In order to directly establish the link between the microstructure, stability, and rheology of bijels, we have completed the integration and upgrade of a Discovery Hybrid Rheometer (DHR, TA Instruments) with a microscopy module and a confocal scanner (Vt-Eye, Visitech International), which will enable confocal imaging of bijels and other soft materials during rheological testing. Our efforts have focused on proper integration of the two systems, characterizing the flow field, and ensuring proper alignment of the optical and flow axes. Our results on the newly integrated system have not yet shown successful formation of bijels directly on a rheometer stage, but efforts are underway to ameliorate this problem. To achieve this, we will equip the rheometer stage with additional heating/cooling elements to provide a more rapid temperature change, while simultaneously continuing our efforts in synthesizing smaller particles with neutral wettability, which should further reduce local deformations to the fluid interface.

3) We have begun a comprehensive study of the unique morphological aspects of bijels and microporous materials derived from them in the context of multiphase transport through porous media. We hypothesize that the uniform pore diameter and continuous pore phase in a bijel results in higher hydraulic permeability, and also helps improve transport kinetics through these porous materials. To test our hypothesis, we have embarked on a comprehensive study to quantify the salient morphological characteristics of microporous bijel-derived materials and their role in mediating fluid flow through the material. The project involves synthesis, 3D scanning, computational geometry, and computational fluid dynamics, which will be performed on bijels and three other classes of commonly used microporous materials (commercially available porous metals, emulsion-templated random porous structures, and inverse opals). Our efforts in the review period have focused on developing the synthesis, computational geometry, and computational fluid dynamics tools, which have been successfully tested on bijels. The results of this study will be submitted for publication during the upcoming period.

Bibliography Type:	Description: (Last Updated: 04/22/2025)
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